



SIMATS Online Programming Lab

JAVA PROGRAMMING

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ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner input = new Scanner(System.in);
5         int num1=input.nextInt();
6         int num2=input.nextInt();
7         System.out.println("sum:"+ (num1+num2));
8         System.out.println("difference:"+ (num1-num2));
9         System.out.println("product:"+ (num1*num2));
10        System.out.println("quotient:"+ (num1/num2));
11        input.close();
12    }
13 }
```

OUTPUT

```
sum:15
difference:-5
product:50
quotient:0
```



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MAX

Write a Java program to find the maximum of three numbers using conditional statements.

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner input=new Scanner(System.in);
5         int a=input.nextInt();
6         int b=input.nextInt();
7         int c=input.nextInt();
8         if(a>=b&&a>=c){
9             System.out.println("max:"+a);
10        }else if(b>=a&&b>=c){
11            System.out.println("max:"+b);
12        }else{
13            System.out.println("max:"+c);
14        }
15        input.close();
16    }
17 }
```

OUTPUT

```
max:9
```



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Leap

LEAP
Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner input=new Scanner(System.in);
5         int year = input.nextInt();
6         if((year%4==0&&year%100!=0)|| (year%400==0)){
7             System.out.println("leap year");
8         }else{
9             System.out.println("not a leap year");
10        }
11        input.close();
12    }
13 }
```

OUTPUT

leap year

2024



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QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI
Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         if(sc.hasNextInt()){
6             int n=sc.nextInt();
7             int first=0,second=1;
8             for (int i=0;i<n;i++){
9                 System.out.print(first+" ");
10                int next = first+second;
11                first=second;
12                second=next;
13            }
14        }
15        sc.close();
16    }
17 }
```

OUTPUT

0 1 1 2 3

INPUT

5



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QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

JavaRunSave

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         if(sc.hasNextInt()){
6             int n = sc.nextInt();
7             long factorial = 1;
8             for(int i=1;i<=n;i++){
9                 factorial*=i;
10            }
11            System.out.println(factorial);
12        }
13        sc.close();
14    }
15 }
```

OUTPUT

120

INPUT

5



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RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
**

PROGRAM EDITOR

JavaRunSave

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int n=5;
6         if(sc.hasNextInt()){
7             n=sc.nextInt();
8         }
9         for(int i=1;i<=n;i++){
10            for(int j=1;j<=i;j++){
11                System.out.print("**");
12            }
13            System.out.println();
14        }
15        sc.close();
16    }
17 }
```

OUTPUT

**
**
**

Your INPUT goes here!



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Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for(int i=0;i<n;i++){a[i]=sc.nextInt();}
8         int key = sc.nextInt(),low=0,high=n-1;
9         while(low<=high){
10             int mid=(low+high)/2;
11             if (a[mid]==key){
12                 System.out.println("Found at index : "+mid);
13                 return;
14             }
15             if(a[mid]<key)low=mid+1;
16             else high=mid-1;
17         }
18         System.out.println("not found");
19     }
20 }
```

OUTPUT

Found at index : 2

5
10 20 30 40 50
30



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RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
*
**


```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=5;
6         if(sc.hasNextInt()){
7             n=sc.nextInt();
8         }
9         for(int i=1;i<=n;i++){
10             for(int j=1;j<=i;j++){
11                 System.out.print("**");
12             }
13             System.out.println();
14         }
15         sc.close();
16     }
17 }
```

OUTPUT

*
**

Your INPUT goes here!

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for(int i=0;i<n;i++){a[i]=sc.nextInt();}
8         int key = sc.nextInt(),low=0,high=n-1;
9         while(low<=high){
10             int mid=(low+high)/2;
11             if (a[mid]==key){
12                 System.out.println("Found at index : "+mid);
13                 return;
14             }
15             if(a[mid]<key)low=mid+1;
16             else high=mid-1;
17         }
18         System.out.println("not found");
19     }
20 }
```

OUTPUT

```
Found at index : 2
```

```
5
10 20 30 40 50
30
```

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         if(sc.hasNextInt()){
6             int a = sc.nextInt();
7             int b = sc.nextInt();
8             while(b!=0){
9                 int temp=b;
10                b=a%b;
11                a=temp;
12            }
13            System.out.println(a);
14        }
15        sc.close();
16    }
17 }
```

OUTPUT

```
6
```

```
12 18
```



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Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=7;
4         if(n<=1){
5             System.out.println("not prime");
6             return;
7         }
8         for(int i=2;i<=n;i++){
9             if(n%i==0){
10                System.out.println("not prime");
11                return;
12            }
13        }
14        System.out.println("Prime");
15    }
16 }
```

OUTPUT

Prime

Your INPUT goes here!



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Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         if(!sc.hasNextInt()) return;
6         int n=sc.nextInt();
7         int[] a=new int[n];
8         for(int i = 0 ; i<n;i++) a[i]=sc.nextInt();
9         for(int i=0;i<n-1;i++){
10            for(int j=i+1;j<n;j++){
11                if(a[j]<a[i]){
12                    int t=a[i];
13                    a[i]=a[j];
14                    a[j]=t;
15                }
16            }
17        }
18        for(int x:a)System.out.print(x+" ");
19    }
20 }
21
```

OUTPUT

10 24 32 45 65

5
32 45 65 24 10



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Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

```
1 public class Main{
2 public static void main(String[] args){
3 int n=5;
4 for(int i=5;i<=10;i++)
5 System.out.println( n + "x" + i + "=" + (n*i));
6 }
7 }
```

OUTPUT

5x5=25
5x6=30
5x7=35

Your INPUT goes here!



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Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

```
1 public class main{
2 static int fact(int n){
3 if(n==0||n==1)
4 return 1;
5 return n*fact(n-1);
6 }
7 public static void main(String[] args){
8 int n=5;
9 System.out.print(fact(n));
10 }
11 }
```

OUTPUT

120

Your INPUT goes here!



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LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**

*

```
1 public class Main{
2     public static void main(String[] args){
3         for(int i=5;i>=1;i--){
4             for(int j=1;j<=i;j++)
5                 System.out.print("*");
6             System.out.println();
7         }
8     }
9 }
```

OUTPUT

**
*

Your INPUT goes here!



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LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int a=12,b=18,lcm;
4         for(lcm=(a>b?a:b);lcm++)
5             if(lcm%a==0&&lcm%b==0)
6                 break;
7         System.out.println(lcm);
8     }
9 }
```

OUTPUT

36

Your INPUT goes here!



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D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=10,b=0,p=1;
4         while(n>0){
5             b=b+(n%2)*p;
6             n/=2;
7             p*=10;
8         }
9         System.out.print(b);
10    }
11 }
```

OUTPUT

1010

Your INPUT goes here!



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B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=1010,d=0,p=1;
4         while(n>0){
5             d=d+(n%10)*p;
6             p*=2;
7             n/=10;
8         }
9         System.out.print(d);
10    }
11 }
```

OUTPUT

10

Your INPUT goes here!

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=1234,sum=0;
4         while(n>0){
5             sum+=n%10;
6             n/=10;
7         }
8         System.out.print(sum);
9     }
10 }
```

Your INPUT goes
here!

OUTPUT

10

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=1234,p=1;
4         while(n>0){
5             p*=n%10;
6             n/=10;
7         }
8         System.out.print(p);
9     }
10 }
```

Your INPUT goes
here!

OUTPUT

24



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Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

```
1 public class Main{
2   public static void main(String[] args){
3     int n=2,p=3,ans=1;
4     for(int i=1;i<=p;i++)
5       ans*=n;
6     System.out.print(ans);
7   }
8 }
```

OUTPUT

8

Your INPUT goes here!



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Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

```
1 public class Main{
2   public static void main(String[] args){
3     double r=5;
4     System.out.print(((4.0/3)*3.14*r*r));
5   }
6 }
```

OUTPUT

523.3333333333334

Your INPUT goes here!

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
1 public class Main{
2     public static void main(String[] args){
3         double r=5;
4         System.out.print((4.0/3)*3.14*r*r*r);
5     }
6 }
```

Your INPUT goes here!

OUTPUT

523.3333333333334

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         String s="madam",r="";
4         for(int i=s.length()-1;i>=0;i--)
5             r+=s.charAt(i);
6         System.out.print(s.equals(r)?"palindrome":"not palindrome");
7     }
8 }
```

Your INPUT goes here!

OUTPUT

palindrome



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Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=134,t=n,s=0;
4         while(n>0){
5             int d=n%10;
6             s+=d*d*d;
7             n/=10;
8         }
9         System.out.print(t==s?"armstrong":"not armstrong");
10    }
11 }
```

OUTPUT

not armstrong

Your INPUT goes here!

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:
1
22
333
4444
55555

```
1 public class Main{
2     public static void main(String[] args){
3         for(int i=1;i<=5;i++){
4             for(int j=1;j<=i;j++){
5                 System.out.print(i);
6                 System.out.println();
7             }
8         }
9     }
```

OUTPUT

1
22
333

Your INPUT goes here!



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Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int n=6,s=0;
4         for(int i=1;i<n;i++)
5             if(n%i==0)
6                 s+=i;
7         System.out.print(s==n?"perfect":"not perfect");
8     }
9 }
```

OUTPUT

perfect

Your INPUT goes here!



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Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int[] a={5,3,7,9,7};
4         int max=a[0];
5         for(int i=1;i<a.length;i++)
6             if(a[i]>max)
7                 max=a[i];
8         System.out.print(max);
9     }
10 }
```

OUTPUT

9

Your INPUT goes here!



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2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
1 public class Main{
2   public static void main(String[] args){
3     int[] a={5,6,7,3,8};
4     int max=Integer.MIN_VALUE,second=Integer.MIN_VALUE;
5     for(int x:a){
6       if(x>max){
7         second=max;
8         max=x;
9       }else if(x > second && x != max){
10        second=x;
11      }
12    }
13    System.out.print(second);
14  }
15 }
```

OUTPUT

7

Your INPUT goes here!



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Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

```
1 public class Main{
2   public static void main(String[] args){
3     int[] a={1,4,6,7,9};
4     int min = a[0];
5     for(int i=1;i<a.length;i++){
6       if(a[i]<min)
7         min=a[i];
8     }
9     System.out.print(min);
10  }
```

OUTPUT

1

Your INPUT goes here!

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int[] a={5,4,3,2,8};
4         int min=Integer.MAX_VALUE,second=Integer.MAX_VALUE;
5         for(int x:a){
6             if(x<min){
7                 second=min;
8                 min=x;
9             }else if(x<second&&x!=min){
10                second=x;
11            }
12        }
13        System.out.print(second);
14    }
15 }
```

Your INPUT goes here!

OUTPUT

3

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int[] a={5,2,7,8,3};
4         for(int i=0;i<a.length-1;i++)
5             for(int j=0;j<a.length-i-1;j++)
6                 if(a[j]>a[j+1]){
7                     int t = a[j];
8                     a[j]=a[j+1];
9                     a[j+1]=t;
10                }
11        for(int x:a)
12            System.out.print(x+" ");
13    }
14 }
```

Your INPUT goes here!

OUTPUT

5 2 7 8 3

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
1 public class Main{
2     public static void main(String[] args){
3         int[] a = {5,3,4,1,2};
4         for(int i=1;i<a.length;i++){
5             int key = a[i],j=i-1;
6             while(j>=0&& a[j]>key){
7                 a[j+1]=a[j];
8                 j--;
9             }
10            a[j+1]=key;
11        }
12        for(int x:a)
13            System.out.print(x+" ");
14    }
15 }
```

OUTPUT

```
1
2
```

Your INPUT goes
here!